

Lee-Sup Kim

professor

基本信息

姓名: Lee-Sup Kim

所属院校: 韩国科学技术院

地区: South Korea

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

学科领域

电子与计算机工程

人工智能

集成电路设计与集成系统

研究领域

计算机体系结构

深度学习与神经网络

低功耗设计与能耗优化

嵌入式系统与硬件设计

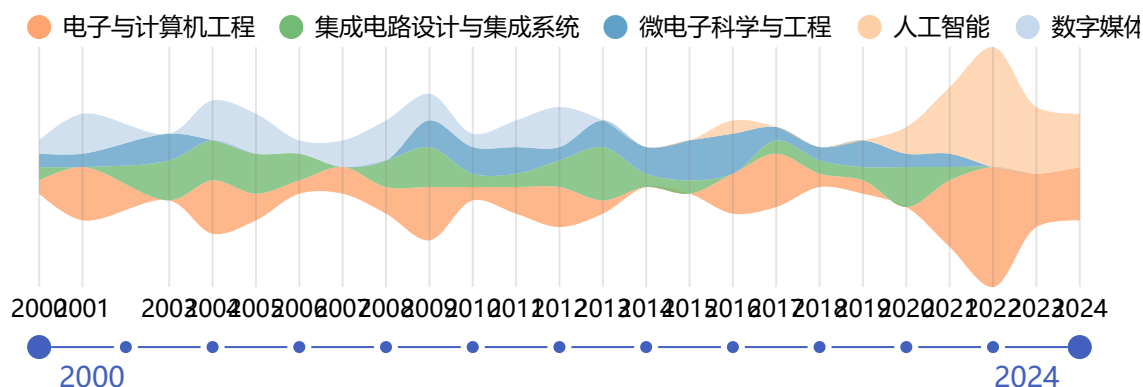
低功耗电路与能量管理

高性能计算与并行处理

个人简介

Lee - Sup Kim is a professor at the School of Electrical Engineering. He holds a B.S. from Seoul National University (1982), an M.S. and a Ph.D. from Stanford University (1986, 1990). He has been a professor at KAIST since 1993 and was a post - doctoral researcher at Toshiba Research Center from 1990 - 1993. His research interests include energy - efficient hardware architecture for deep learning and in - memory computing for neural networks. He has major achievements like the 1.42TOPS/W deep convolutional neural network recognition processor. With 268 research outputs, including articles and conference contributions, his work is in areas such as electric power utilization, deep neural networks, and energy efficiency.

近年发表的论文



ToEx: Accelerating Generation Stage of Transformer-Based Language Models via Token-Adaptive Early Exit

IEEE TRANSACTIONS ON COMPUTERS

Kang, Myeonggu; Park, Junyoung; Shin, Hyein; Shin, Jaekang; Kim, Lee-Sup

被引用次数: 0

2024

ADC-Free ReRAM-Based In-Situ Accelerator for Energy-Efficient Binary Neural Networks

IEEE TRANSACTIONS ON COMPUTERS

Kim, Hyeonuk; Jung, Youngbeom; Kim, Lee-Sup

被引用次数: 1

2024

Accelerating Deep Reinforcement Learning via Phase-Level Parallelism for Robotics Applications

IEEE COMPUTER ARCHITECTURE LETTERS

Kim, Yang-Gon; Han, Yun-Ki; Shin, Jae-Kang; Kim, Jun-Kyum; Kim, Lee-Sup

被引用次数: 0

2024

Token-Picker: Accelerating Attention in Text Generation with Minimized Memory Transfer via Probability Estimation

Design Automation Conference

Junyoung Park; Myeonggu Kang; Yunki Han; Yang-Gon Kim; Jaekang Shin; Lee-Sup Kim

被引用次数: 0

2024

MGen: A Framework for Energy-Efficient In-ReRAM Acceleration of Multi-Task BERT

IEEE TRANSACTIONS ON COMPUTERS

Kang, Myeonggu; Shin, Hyein; Kim, Junkyum; Kim, Lee-Sup

被引用次数: 1

2023

Fault-Free: A Framework for Analysis and Mitigation of Stuck-at-Fault on Realistic ReRAM-Based DNN Accelerators

IEEE TRANSACTIONS ON COMPUTERS

Shin, Hyein; Kang, Myeonggu; Kim, Lee-Sup

被引用次数: 1

2023

Accelerating On-Device DNN Training Workloads via Runtime Convergence Monitor

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Choi, Seungkyu; Shin, Jaekang; Kim, Lee-Sup

被引用次数: 0

2023

Energy-Efficient CNN Personalized Training by Adaptive Data Reformation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Jung, Youngbeom; Kim, Hyeonuk; Choi, Seungkyu; Shin, Jaekang; Kim, Lee-Sup

被引用次数: 3

2023

OptimStore: In-Storage Optimization of Large Scale DNNs with On-Die Processing

IEEE International Symposium on High Performance Computer Architecture

Junkyum Kim; Myeonggu Kang; Yunki Han; Yang-Gon Kim; Lee-Sup Kim

被引用次数: 0

2023

EGCN: An Efficient GCN Accelerator for Minimizing Off-Chip Memory Access

IEEE TRANSACTIONS ON COMPUTERS

Han, Yunki; Park, Kangkyu; Jung, Youngbeom; Kim, Lee-Sup

被引用次数: 4

2022

S-FLASH: A NAND Flash-Based Deep Neural Network Accelerator Exploiting Bit-Level Sparsity

IEEE TRANSACTIONS ON COMPUTERS

Kang, Myeonggu; Kim, Hyeonuk; Shin, Hyein; Sim, Jaehyeong; Kim, Kyeonghan; Kim, Lee-Sup

被引用次数: 8

2022

A Deep Neural Network Training Architecture With Inference-Aware Heterogeneous Data-Type

IEEE TRANSACTIONS ON COMPUTERS

Choi, Seungkyu; Shin, Jaekang; Kim, Lee-Sup

被引用次数: 4

2022

Rare Computing: Removing Redundant Multiplications From Sparse and Repetitive Data in Deep Neural Networks

IEEE TRANSACTIONS ON COMPUTERS

Park, Kangkyu; Choi, Seungkyu; Choi, Yeongjae; Kim, Lee-Sup

被引用次数: 0

2022

A Framework for Accelerating Transformer-Based Language Model on ReRAM-Based Architecture

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Kang, Myeonggu; Shin, Hyein; Kim, Lee-Sup

被引用次数: 6

2022

Quantization-Error-Robust Deep Neural Network for Embedded Accelerators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jung, Youngbeom; Kim, Hyeonuk; Choi, Yeongjae; Kim, Lee-Sup

被引用次数: 2

2022

Algorithm/Architecture Co-Design for Energy-Efficient Acceleration of Multi-Task DNN

PROCEEDINGS OF THE 59TH ACM/IEEE DESIGN AUTOMATION CONFERENCE, DAC 2022

Shin, Jaekang; Choi, Seungkyu; Ra, Jongwoo; Kim, Lee -Sup

被引用次数: 1

2022

Re2fresh: A Framework for Mitigating Read Disturbance in ReRAM-based DNN Accelerators

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Shin, Hyein; Kang, Myeonggu; Kim, Lee-Sup

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2022

Algorithm/architecture co-design for energy-efficient acceleration of multi-task DNN

Design Automation Conference

Jaekang Shin; Seungkyu Choi; Jongwoo Ra; Lee-Sup Kim

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2022

Amnesiac DRAM: A Proactive Defense Mechanism Against Cold Boot Attacks

IEEE TRANSACTIONS ON COMPUTERS

Seol, Hoseok; Kim, Minhye; Kim, Taesoo; Kim, Yongdae; Kim, Lee-Sup

被引用次数: 4

2021

Fault-free: A Fault-resilient Deep Neural Network Accelerator based on Realistic ReRAM Devices

Design Automation Conference

Hyein Shin; Myeonggu Kang; Lee-Sup Kim

被引用次数: 0

2021

Optimizing ADC Utilization through Value-Aware Bypass in ReRAM-based DNN Accelerator

Design Automation Conference

HanCheon Yun; Hyein Shin; Myeonggu Kang; Lee-Sup Kim

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2021

Fault-free: A Fault-resilient Deep Neural Network Accelerator based on Realistic ReRAM Devices

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Shin, Hyein; Kang, Myeonggu; Kim, Lee-Sup

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2021

Optimizing ADC Utilization through Value-Aware Bypass in ReRAM-based DNN Accelerator

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Yun, HanCheon; Shin, Hyein; Kang, Myeonggu; Kim, Lee-Sup

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2021

A Convergence Monitoring Method for DNN Training of On-Device Task Adaptation

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Choi, Seungkyu; Shin, Jaekang; Kim, Lee-Sup

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2021

A Framework for Area-efficient Multi-task BERT Execution on ReRAM-based Accelerators

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Kang, Myeonggu; Shin, Hyein; Shin, Jaekang; Kim, Lee-Sup

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2021

Deferred Dropout: An Algorithm-Hardware Co-Design DNN Training Method Provisioning Consistent High Activation Sparsity

2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Park, Kangkyu; Han, Yunki; Kim, Lee-Sup

被引用次数: 2

2021

An Energy-Efficient Deep Convolutional Neural Network Training Accelerator for In Situ Personalization on Smart Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Seungkyu; Sim, Jaehyeong; Kang, Myeonggu; Choi, Yeongjae; Kim, Hyeonuk; Kim, Lee-Sup

被引用次数: 34

2020

CREMON: Cryptography Embedded on the Convolutional Neural Network Accelerator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Yeongjae; Sim, Jaehyeong; Kim, Lee-Sup

被引用次数: 8

2020

An Energy-Efficient Deep Convolutional Neural Network Inference Processor With Enhanced Output Stationary Dataflow in 65-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Sim, Jaehyeong; Lee, Somin; Kim, Lee-Sup

被引用次数: 44

2020

A Pragmatic Approach to On-device Incremental Learning System with Selective Weight Updates

Design Automation Conference

Jaekang Shin; Seungkyu Choi; Yeongjae Choi; Lee-Sup Kim

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2020

DC-PCM: Mitigating PCM Write Disturbance with Low Performance Overhead by Using Detection Cells

IEEE TRANSACTIONS ON COMPUTERS

Choi, Jungwhan; Jang, Jaemin; Kim, Lee-Sup

被引用次数: 7

2019

Sparse-Insertion Write Cache to Mitigate Write Disturbance Errors in Phase Change Memory

IEEE TRANSACTIONS ON COMPUTERS

Jang, Jaemin; Shin, Wongyu; Choi, JungWhan; Kim, Yongju; Kim, Lee-Sup

被引用次数: 5

2019

A 12 Gb/s 1.59 mW/Gb/s Input-Data-Jitter-Tolerant Injection-Type CDR With Super-Harmonic Injection-Locking in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jung, Chongsoo; Lee, Dongil; Kim, Yong-Hun; Lee, Daewoong; Kim, Lee-Sup

被引用次数: 4

2019

A 0.9-V 12-Gb/s Two-FIR Tap Direct DFE With Feedback-Signal Common-Mode Control

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Daewoong; Lee, Dongil; Kim, Yong-Hun; Kim, Lee-Sup

被引用次数: 10

2019

Elaborate Refresh: A Fine Granularity Retention Management for Deep Submicron DRAMs

IEEE TRANSACTIONS ON COMPUTERS

Seol, Hoseok; Shin, Wongyu; Jang, Jaemin; Choi, Jungwhan; Lee, Hakseung; Kim, Lee-Sup

被引用次数: 4

2018

A 10-Gb/s Reference-Less Baud-Rate CDR for Low Power Consumption With the Direct Feedback Method

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Yong-Hun; Lee, Dongil; Lee, Daewoong; Kim, Lee-Sup

被引用次数: 2

2018

A 0.65-V, 11.2-Gb/s Power Noise Tolerant Source-Synchronous Injection-Locked Receiver With Direct DTLB DFE

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Dongil; Kim, Yong-Hun; Lee, Daewoong; Kim, Lee-Sup

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2018

Bank-Group Level Parallelism

IEEE TRANSACTIONS ON COMPUTERS

Shin, Wongyu; Jang, Jaemin; Choi, Jungwhan; Suh, Jinwoong; Kim, Lee-Sup

被引用次数: 2

2017

Rank-Level Parallelism in DRAM

IEEE TRANSACTIONS ON COMPUTERS

Shin, Wongyu; Jang, Jaemin; Choi, Jungwhan; Suh, Jinwoong; Kwon, Yongkee; Moon, Youngsuk; Kim, Lee-Sup

被引用次数: 4

2017

Refresh-Aware Write Recovery Memory Controller

IEEE TRANSACTIONS ON COMPUTERS

Jang, Jaemin; Shin, Wongyu; Choi, Jungwhan; Suh, Jinwoong; Kwon, Yongkee; Kim, Yongju; Kim, Lee-Sup

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2017

An Input Data and Power Noise Inducing Clock Jitter Tolerant Reference-Less Digital CDR for LCD Intra-Panel Interface

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Yong-Hun; Lee, Taeho; Jeon, Hyun-Kyu; Lee, Dongil; Kim, Lee-Sup

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2017

Energy-Efficient Design of Processing Element for Convolutional Neural Network

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Yeongjae; Bae, Dongmyung; Sim, Jaehyeong; Choi, Seungkyu; Kim, Minhye; Kim, Lee-Sup

被引用次数: 26

2017

In-DRAM Data Initialization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Seol, Hoseok; Shin, Wongyu; Jang, Jaemin; Choi, Jungwhan; Suh, Jinwoong; Kim, Lee-Sup

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2017

A 5-Gb/s Digital Clock and Data Recovery Circuit With Reduced DCO Supply Noise Sensitivity Utilizing Coupling Network

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Taeho; Kim, Yong-Hun; Kim, Lee-Sup

被引用次数: 9

2017

DRAM-Latency Optimization Inspired by Relationship between Row-Access Time and Refresh Timing

IEEE TRANSACTIONS ON COMPUTERS

Shin, Wongyu; Choi, Jungwhan; Jang, Jaemin; Suh, Jinwoong; Moon, Youngsuk; Kwon, Yongkee; Kim, Lee-Sup

被引用次数: 8

2016

Q-DRAM: Quick-Access DRAM with Decoupled Restoring from Row-Activation

IEEE TRANSACTIONS ON COMPUTERS

Shin, Wongyu; Choi, Jungwhan; Jang, Jaemin; Suh, Jinwoong; Kwon, Yongkee; Moon, Youngsuk; Kim, Hongsik; Kim, Lee-Sup

被引用次数: 8

2016

A Vision Processor With a Unified Interest-Point Detection and Matching Hardware for Accelerating a Stereo-Matching Algorithm

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Park, Jun-Seok; Kim, Hyo-Eun; Kim, Hong-Yun; Lee, Jaeyoung; Kim, Lee-Sup

被引用次数: 12

2016

A 21%-Jitter-Improved Self-Aligned Dividerless Injection-Locked PLL With a VCO Control Voltage Ripple-Compensated Phase Detector

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Dongil; Lee, Taeho; Kim, Young-Ju; Kim, Lee-Sup

被引用次数: 7

2016

A 10-Gb/s 0.71-pJ/bit Forwarded-Clock Receiver Tolerant to High-Frequency Jitter in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chung, Sang-Hye; Kim, Young-Ju; Kim, Yong-Hun; Kim, Lee-Sup

被引用次数: 5

2016

A 21-Gbit/s 1.63-pJ/bit Adaptive CTLE and One-Tap DFE With Single Loop Spectrum Balancing Method

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Yong-Hun; Kim, Young-Ju; Lee, Taeho; Kim, Lee-Sup

被引用次数: 30

2016

Hybrid Temperature Sensor Network for Area-Efficient On-Chip Thermal Map Sensing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Paek, Seungwook; Shin, Wongyu; Lee, Jaeyoung; Kim, Hyo-Eun; Park, Jun-Seok; Kim, Lee-Sup

被引用次数: 9

2015

A 9.6 Gb/s 0.96 mW/Gb/s Forwarded Clock Receiver With High Jitter Tolerance Using Mixing Cell Integrated Injection-Locked Oscillator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Young-Ju; Chung, Sang-Hye; Ha, Kyung-Soo; Bae, Seung-Jun; Kim, Lee-Sup

被引用次数: 4

2015

A 9.6-Gb/s 1.22-mW/Gb/s Data-Jitter Mixing Forwarded-Clock Receiver in 65-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chung, Sang-Hye; Kim, Lee-Sup

被引用次数: 3

2015

An 11.5 Gb/s 1/4th Baud-Rate CTLE and Two-Tap DFE With Boosted High Frequency Gain in 110-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Yong-Hun; Kim, Young-Ju; Lee, Tae-Ho; Kim, Lee-Sup

被引用次数: 9

2015

A 5 Gbps 1.6 mW/G bps/CH Adaptive Crosstalk Cancellation Scheme With Reference-less Digital Calibration and Switched Termination Resistors for Single-Ended Parallel Interface

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Hwang, Kyu-Dong; Kim, Lee-Sup

被引用次数: 6

2014

A Quarter-Rate Forwarded Clock Receiver Based on ILO With Low Jitter Tracking Bandwidth Variation Using Phase Shifting Phenomenon in 65 nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Young-Ju; Chung, Sang-Hye; Kim, Lee-Sup

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2014

A Forwarded-Clock Receiver With Constant and Wide-Range Jitter-Tracking Bandwidth

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chung, Sang-Hye; Kim, Young-Ju; Ha, Kyung-Soo; Bae, Seung-Jun; Lee, Jung-Bae; Kim, Lee-Sup

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2014

PowerField: A Probabilistic Approach for Temperature-to-Power Conversion Based on Markov Random Field Theory

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Paek, Seungwook; Shin, Wongyu; Sim, Jaehyeong; Kim, Lee-Sup

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2013

A 1 mJ/Frame Unified Media Application Processor With Dynamic Analog-Digital Mode Reconfiguration for Embedded 3D-Media Contents Processing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyo-Eun; Park, Jun-Seok; Yoon, Jae-Sung; Kim, Seok-Hoon; Kim, Lee-Sup

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2013

A 182mW 94.3 f/s in Full HD Pattern-Matching Based Image Recognition Accelerator for an Embedded Vision System in 0.13- μ m CMOS Technology

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Park, Jun-Seok; Kim, Hyo-Eun; Kim, Lee-Sup

被引用次数: 23

2013

A Reconfigurable SIMT Processor for Mobile Ray Tracing With Contention Reduction in Shared Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Hong-Yun; Kim, Young-Jun; Oh, Jie-Hwan; Kim, Lee-Sup

被引用次数: 17

2013

A 6.5-Gb/s 1-mW/Gb/s/CH Simple Capacitive Crosstalk Compensator in a 130-nm Process

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Hwang, Kyu-Dong; Kim, Lee-Sup

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2013

A LOG-Induced SSN-Tolerant Transceiver for On-Chip Interconnects in COG-Packaged Source Driver IC for TFT-LCD

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Won-Young; Oh, Jiehwan; Kim, Lee-Sup

被引用次数: 1

2013

A Unified Graphics and Vision Processor With a 0.89 μ W/fps Pose Estimation Engine for Augmented Reality

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yoon, Jae-Sung; Kim, Jeong-Hyun; Kim, Hyo-Eun; Lee, Won-Young; Kim, Seok-Hoon; Chung, Kyusik; Park, Jun-Seok; Kim, Lee-Sup

被引用次数: 19

2013

M RTP: Mobile Ray Tracing Processor With Reconfigurable Stream Multi-Processors for High Datapath Utilization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hong-Yun; Kim, Young-Jun; Kim, Lee-Sup

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2012

A Reconfigurable Heterogeneous Multimedia Processor for IC-Stacking on Si-Interposer

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Kim, Hyo-Eun; Yoon, Jae-Sung; Hwang, Kyu-Dong; Kim, Young-Jun; Park, Jun-Seok; Kim, Lee-Sup

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2012

A 5.4/2.7/1.62-Gb/s Receiver for DisplayPort Version 1.2 With Multi-Rate Operation Scheme

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Won-Young; Hwang, Kyu-Dong; Kim, Lee-Sup

被引用次数: 16

2012

A 5.4-Gb/s Clock and Data Recovery Circuit Using Seamless Loop Transition Scheme With Minimal Phase Noise Degradation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Won-Young; Kim, Lee-Sup

被引用次数: 7

2012

Homogeneous Stream Processors With Embedded Special Function Units for High-Utilization Programmable Shaders

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Young-Jun; Kim, Hyo-Eun; Kim, Seok-Hoon; Park, Jun-Seok; Paek, Seungwook; Kim, Lee-Sup

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2012

A Mobile 3-D Display Processor With A Bandwidth-Saving Subdivider

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Seok-Hoon; Yoon, Sung-Eui; Chung, Sang-Hye; Kim, Young-Jun; Kim, Hong-Yun; Chung, Kyusik; Kim, Lee-Sup

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2012

A Memory-Efficient Unified Early Z-Test

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS

Kim, Hong-Yun; Yu, Chang-Hyo; Kim, Lee-Sup

被引用次数: 4

2011

A 100 MHz-to-1 GHz Fast-Lock Synchronous Clock Generator With DCC for Mobile Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Mi-Jo; Kim, Lee-Sup

被引用次数: 9

2011

A Spread Spectrum Clock Generator for DisplayPort Main Link

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Won-Young; Kim, Lee-Sup

被引用次数: 15

2011

A Dual-Shader 3-D Graphics Processor With Fast 4-D Vector Inner Product Units and Power-Aware Texture Cache

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yoon, Jae-Sung; Yu, Chang-Hyo; Kim, Donghyun; Kim, Lee-Sup

被引用次数: 9

2011

Adaptive gain push-pull shunt regulation in digital circuits

ELECTRONICS LETTERS

Jeong, H. -S.; Kim, L. -S.

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2011

A 116 fps/74 mW Heterogeneous 3D-Media Processor for 3-D Display Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Seok-Hoon; Kim, Hong-Yun; Kim, Young-Jun; Chung, Kyusik; Kim, Donghyun; Kim, Lee-Sup

被引用次数: 11

2010

A 5-Gb/s/pin Transceiver for DDR Memory Interface With a Crosstalk Suppression Scheme (vol 44, pg 2222, 2009)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Kwang-Il; Kim, Lee-Sup; Park, Kwang-Il; Jun, Young-Hyun; Choi, Joo Sun; Kim, Kinam

被引用次数: 0

2010

A Data-Pattern-Tolerant Adaptive Equalizer Using the Spectrum Balancing Method

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Joo, Hye-Yoon; Kim, Lee-Sup

被引用次数: 18

2010

Infinite Etch Selectivity during Etching of SiON with an Extreme Ultraviolet Resist Pattern in Dual-Frequency Capacitively Coupled Plasmas

JOURNAL OF THE ELECTROCHEMICAL SOCIETY

Kwon, B. S.; Kim, J. S.; Lee, N. -E.; Lee, S. K.

被引用次数: 3

2010

Three-level differential buffer for increasing noise margin in pseudo-differential signalling

ELECTRONICS LETTERS

Ha, K. -S.; Kim, L. -S.; Bae, S. -J.; Park, K. -I.; Choi, J. S.; Jun, Y. -H.

被引用次数: 0

2010

A Floating-Point Unit for 4D Vector Inner Product with Reduced Latency

IEEE TRANSACTIONS ON COMPUTERS

Kim, Donghyun; Kim, Lee-Sup

被引用次数: 22

2009

Shader-based tessellation to save memory bandwidth in a mobile multimedia processor

COMPUTERS & GRAPHICS-UK

Chung, Kyusik; Yu, Chang-Hyo; Kim, Donghyun; Kim, Lee-Sup

被引用次数: 9

2009

A 0.13- μ m CMOS 6 Gb/s/pin Memory Transceiver Using Pseudo-Differential Signaling for Removing Common-Mode Noise Due to SSN

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ha, Kyung-Soo; Kim, Lee-Sup; Bae, Seung-Jun; Park, Kwang-Il; Choi, Joo Sun; Jun, Young-Hyun; Kim, Kinam

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2009

A 5-Gb/s/pin Transceiver for DDR Memory Interface With a Crosstalk Suppression Scheme

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Kwang-Il; Kim, Lee-Sup; Park, Kwang-Il; Jun, Young-Hyun; Choi, Joo Sun; Kim, Kinam

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2009

A DLL With Jitter Reduction Techniques and Quadrature Phase Generation for DRAM Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Byung-Guk; Kim, Lee-Sup; Park, Kwang-Il; Jun, Young-Hyun; Cho, Soo-In

被引用次数: 14

2009

A 186-Mvertices/s 161-mW Floating-Point Vertex Processor With Optimized Datapath and Vertex Caches

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yu, Chang-Hyo; Chung, Kyusik; Kim, Donghyun; Kim, Seok-Hoon; Kim, Lee-Sup

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2009

Area-efficient special function unit for mobile vertex processors

ELECTRONICS LETTERS

Chung, K.; Kim, L. -S.

被引用次数: 4

2009

Memory bandwidth saving by hardware tessellation with vertex shader

ELECTRONICS LETTERS

Chung, K.; Yu, C. -H.; Kim, D.; Kim, L. -S.

被引用次数: 0

2009

Area-efficient pixel rasterization and texture coordinate interpolation

COMPUTERS & GRAPHICS-UK

Kim, Donghyun; Kim, Lee-Sup

被引用次数: 6

2008

Noise robust motion refinement for motion compensated noise reduction

IEICE TRANSACTIONS ON INFORMATION AND SYSTEMS

Kim, Jong-Sun; Kim, Lee-Sup

被引用次数: 2

2008

A 36 fps SXGA 3-D display processor embedding a programmable 3-D graphics rendering engine

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Seok-Hoon; Yoon, Jae-Sung; Yu, Chang-Hyo; Kim, Donghyun; Chung, Kyusik; Lim, Han Shin; Lee, Yun-Gu; Park, HyunWook; Ra, Jong Beom; Kim, Lee-Sup

被引用次数: 4

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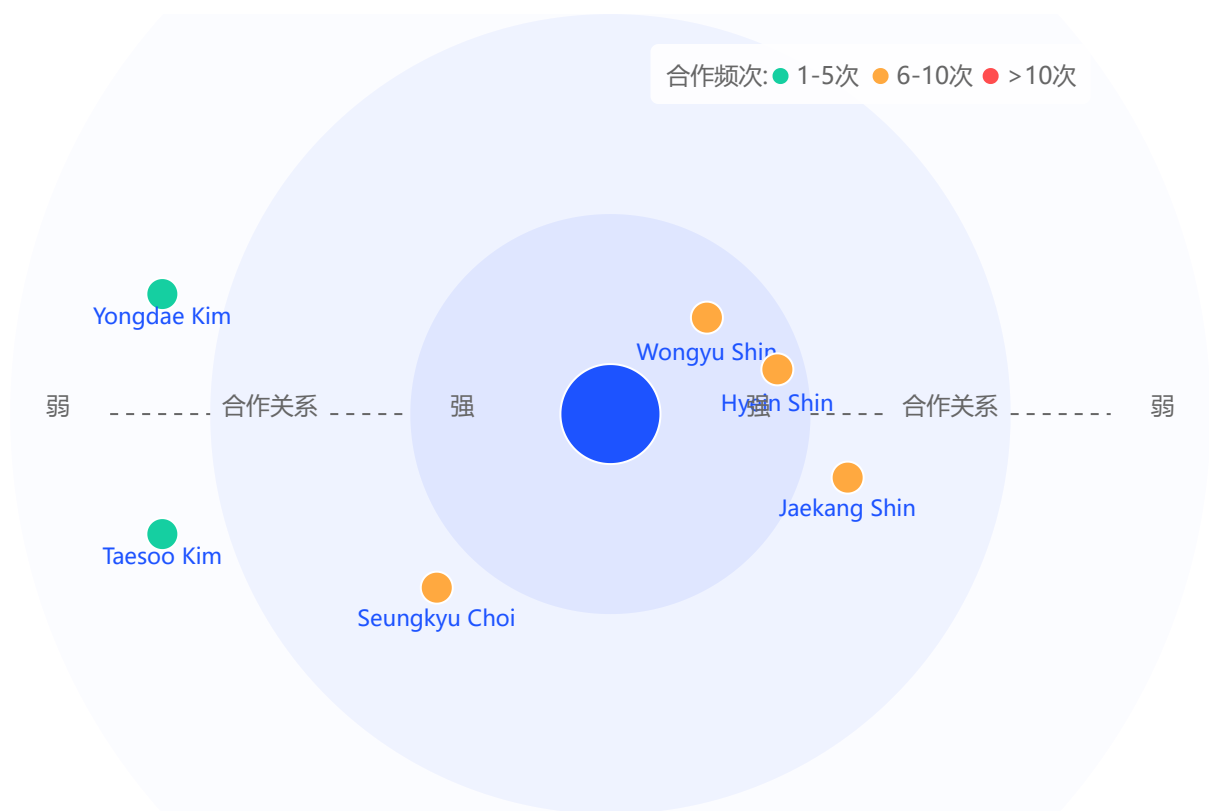
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